

AI501/DS843 Mockup Exam

June 202X

Rules

The exam consists of two parts: a multiple choice section with 30 questions, and a set of open-ended exercises. You will have a total of **3 hours** to complete both parts of the exam, and you may choose the order in which to answer the questions. The suggestion is to start with the multiple choice question to then move to the open-ended questions.

Each correct answer in the multiple choice section gives **+3 points**, while each incorrect answer results in a penalty of **-1 point**. You will have to reach a score of 60% (54 points out of 90) to pass the exam.

For the open-ended exercises, please read carefully the text. For every exercise, provide an answer. Please remember to justify why the results hold by writing the logical flow that lead you to the answers. Note that every exercise have more points/questions. Points can be answered individually (no need to answer all the points of an exercise). To pass the exam, a perfect answer for all questions is not needed.

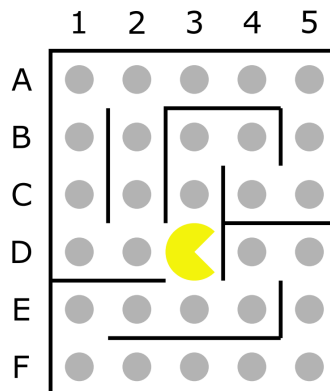
Exercises - Open Questions

1 Friendly Yellow Guy

Friendly Yellow Guy lives in the grid environment shown in the figure below. At each time step, he can move from his current position to any adjacent position, as long as there is no wall between them.

His purpose in life is to eat all the energy pills in his environment – in particular, if he moves to a position where there is a pill, he will always eat it.

Your task is to help Friendly Yellow Guy to complete his task in the shortest number of moves.



- (a) Classify this problem according to the following dimensions: single/multi-agent; fully/partially observable; deterministic/stochastic; discrete/continuous.
- (b) Formulate this problem precisely as a search problem: describe the state space, the initial state, the actions, the transition model, the goal test, and the cost function.
- (c) Describe in your own words or pseudocode the deep-first search algorithm.
- (d) Assume that the possible moves for Friendly Yellow Guy are always tried in the order Left, Up, Right, Down. Write the position of Friendly Yellow Guy in the nodes generated by depth-first search until the first repeated state is reached.
- (e) Consider the heuristic that assigns to each state the number of energy pills left. Explain briefly why this heuristic is admissible. With the same ordering of moves as given before and no check for duplicate states, why is greedy best-first search unable to find a solution to the problem?

2 Slippery Day

Consider the following information.

- If it is sunny, then the grass is dry.
 - The grass is wet if its raining.
 - If the grass is wet and I am wearing sandals, then it is slippery.
 - My feet are cold if I am wearing sandals and its raining.
 - Today, it is raining and I am wearing sandals.
- (a) Write down logical representations for these sentences, suitable for use with forward and backwards chaining. Use a suggestive vocabulary.
- (b) Apply forward chaining to show that it is slippery today. You only need to show the successful steps – i.e., you can assume that you always choose the most helpful formulas.
- (c) Describe in your own words or pseudocode the backward chaining algorithm.
- (d) Apply backward chaining to show that it is slippery. You only need to show the successful steps – i.e., you can assume that you always choose the most helpful formulas.
- (e) Convert your logical representation to conjunctive normal form (CNF) and use resolution to show that it is slippery today.